

L KEVIN DANIEL

Computer Science Student | Software Engineer | Systems Builder
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EDUCATION

Vellore Institute of Technology, Chennai — B.Tech, Computer Science and Engineering

CGPA: **8.69/10** | Current semester: 3

EXPERIENCE

Technical Director — 180 Degrees Consulting, VIT Chennai

- Lead technical initiatives for a student consulting organization, with ownership across software systems, web infrastructure, and technical solutions.
- Drive development and maintenance of internal technology and digital platforms supporting organizational workflows.
- Work across modern web and cloud tooling, translating organizational requirements into maintainable technical solutions.

SELECTED PROJECTS

Synapse — Real-Time Collaborative Knowledge Platform

- Built a cloud-native collaborative document platform using **Next.js**, **TipTap**, **Socket.IO**, **Redis**, and **AWS**, combining rich-text editing with AI-driven knowledge management.
- Engineered a custom **Operational Transformation** layer for concurrent document editing with live cursors and user presence.
- Designed an incrementally updating **personal knowledge graph** that extracts concepts from documents and connects related knowledge across a workspace.
- Integrated AI-generated **summaries**, **flashcards**, and **quizzes** directly from user-authored notes.
- Developing the system toward an academic systems research paper; AWS infrastructure is evolving from **EC2** toward **ECS Fargate** with Redis-backed horizontal scaling for the real-time layer.

MCP — Model Context Protocol Server & Developer Infrastructure

- Implemented a self-hosted **JSON-RPC 2.0 MCP endpoint** with protocol methods including initialization, ping, tool discovery, and tool execution.
- Built a React Flow dashboard for visualizing MCP interactions and implemented tools for filesystem operations and headless browser fetching.
- Developed with **Next.js**, **TypeScript**, **Tailwind CSS**, **Framer Motion**, **Puppeteer**, and **Chromium**, including path-traversal safeguards around the filesystem sandbox.

Quantum Neural Network vs Classical MLP — Experimental ML Research

- Built a comparative binary-classification experiment using **PennyLane**, **PyTorch**, and **scikit-learn** on a noisy two-moons dataset.
- Implemented a variational quantum circuit with angle encoding and entangling layers alongside a classical MLP baseline, with training-loss, accuracy, and ROC-AUC visualization.

180DC Admin Platform — Cloud/Web Infrastructure

- Developed a production-oriented monorepo using **Turborepo** and **Cloudflare Workers**, with separate public API, admin API, and job-processing applications.

TECHNICAL SKILLS

Languages: C++, C, Python, Java, JavaScript, TypeScript, SQL

Web: Next.js, React, Tailwind CSS, TipTap

Systems: Operational Transformation, Real-Time Systems, Socket.IO, Distributed Systems, Concurrency

AI/ML: PyTorch, PennyLane, scikit-learn, Machine Learning, Quantum Machine Learning

Cloud & Infrastructure: AWS, EC2, ECS Fargate, Redis, Cloudflare Workers, Turborepo

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Theory of Computation

Tools: Git, GitHub

RESEARCH & INTERESTS

Systems Research: Real-time collaboration, distributed systems, consistency, horizontal scaling, cloud-native architecture.

AI/ML Research: Quantum machine learning, knowledge graphs, AI-assisted learning systems.

LINKS

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